

User Authentication & Role Management

AI-Native CRM Platform — Module README

Overview

This module handles user authentication, authorization, role management, profile management, and audit logging for the AI-Native CRM platform.

Tech Stack

Frontend: React + TypeScript

Backend: Spring Boot

Database: PostgreSQL

Authentication: JWT

ORM: Hibernate / JPA

Features

1. User Authentication

Login using email and password. Backend validates credentials against PostgreSQL, generates a JWT token after successful authentication, and supports Remember Me.

2. Role-Based Access Control

ADMIN — Full access; EMPLOYEE — Edit-level access; CLIENT — View-level access. Backend endpoints are protected based on the user's role.

3. User Management

Admin can view, create, update, activate/deactivate, and delete users.

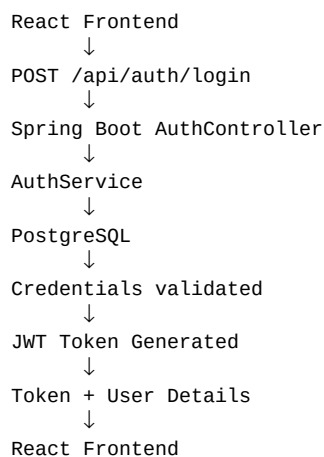
4. Profile Management

Authenticated users can view their profile, update profile information, and change their password.

5. Audit Logging

Important user activities are recorded in the audit_logs table, including login activity, user management actions, IP address, timestamp, user information, and action details.

Authentication Flow



Important API Endpoints

Authentication

POST /api/auth/login

User Profile

GET /api/users/me

PUT /api/users/me

PUT /api/users/me/password

Admin User Management

```
GET    /api/admin/users
GET    /api/admin/users/{id}
POST   /api/admin/users
PUT    /api/admin/users/{id}
PATCH /api/admin/users/{id}/status
DELETE /api/admin/users/{id}
```

Default Admin Account

Email: admin@ainativecrm.com

Password: Admin@1234

Role: ADMIN

Change/remove development credentials before production deployment.

Frontend Structure

```
frontend/
  ■■■ src/
    ■■■ pages/auth/Login.tsx
    ■■■ context/AuthContext.tsx
    ■■■ services/auth.service.ts
    ■■■ hooks/useAuth.ts
    ■■■ types/auth.ts
```

Backend Structure

```
backend/
  ■■■ src/main/java/
    ■■■ controller/
      ■ ■■■ AuthController.java
      ■ ■■■ AdminUserController.java
      ■ ■■■ UserController.java
      ■ ■■■ AuditLogController.java
    ■■■ security/
      ■ ■■■ JwtAuthenticationFilter.java
      ■ ■■■ JwtTokenProvider.java
      ■ ■■■ CustomUserDetailsService.java
    ■■■ service/
      ■ ■■■ AuthService.java
      ■ ■■■ UserService.java
    ■■■ model/
    ■■■ repository/
```

Local Development

Backend

```
cd "C:\Users\Nivetha S\Downloads\ai native crm\backend"
.\gradlew.bat bootRun
```

Backend: <http://localhost:8080>

Frontend

```
cd "C:\Users\Nivetha S\Downloads\ai native crm\frontend"
npm.cmd run dev
```

Frontend: <http://localhost:5173>

Current Integration Status

- ✓ React login UI
- ✓ Spring Boot login API
- ✓ PostgreSQL user authentication
- ✓ JWT authentication
- ✓ Role-based authorization
- ✓ Admin user management APIs
- ✓ Profile APIs

- ✓ Password change API
- ✓ Audit logging
- ✓ Frontend ↔ Backend authentication integration
- ✓ PostgreSQL integration